Cover Letter — Applied Ergonomics submission

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Date: [submission date]

To the Editorial Office,
Applied Ergonomics
Elsevier Ltd.

Dear Editor-in-Chief,

On behalf of my co-author and myself, I am pleased to submit for your consideration the manuscript **"Task-calibrated Operational Performance Indicators for aviation and unmanned aircraft system operators: a biomathematical framework integrating SAFTE fatigue, heart-rate variability, and cognitive-load theory, with open-source reference implementation"** as a Research Article for *Applied Ergonomics*.

What the paper does

The manuscript introduces the **Operational Performance Indicator (OPI) framework** — a task-calibrated composite readiness index that integrates SAFTE-style fatigue effectiveness, heart-rate-variability-derived autonomic markers, task-specific cognitive-load modifiers grounded in Multiple Resource Theory, and environmental/operational modifiers into a single per-task interpretable score. The framework specifies per-task weight profiles and thresholds for **ten manned-aviation task categories** (instrument flight, night-vision operations, helmet-mounted-display flying, high-density air-traffic control, critical and non-critical emergencies, test-pilot operations, carrier landing, weapons delivery, and new-platform testing) and **seven UAS/teleoperator categories** (ISR, strike, SAR/CSAR, autonomous-swarm supervisory control, contested-environment operations, ground-robot teleoperation, and subsea/long-latency teleoperation). A Warm-type vigilance-decrement function and a Chen-type logarithmic control-latency penalty are included for the UAS subset.

The reference implementation is open-source under the MIT license at `https://github.com/strike`
and is delivered through a Next.js client over a FastAPI orchestration layer and a shared Python biomathematical backend. Engineering verification is documented across the fusion pathway. A single 128-minute HRV recording is used as an illustrative framework-instantiation worked example across three task hypotheses.

Why *Applied Ergonomics*

Applied Ergonomics is the natural venue for this contribution for three reasons. First, the paper's central contribution is **methodological** — a theoretically grounded framework for operator-state integration, not a single-study empirical finding — and *Applied Ergonomics* explicitly welcomes methodology papers bearing on the design, planning, and management of technical and social systems in aviation and military contexts. Second, recent precedent (Berthon et al., *Applied Ergonomics* 129, 2025, 104599) confirms that multi-dimensional physiological workload studies in aviation contexts are in scope and that HRV-inclusive composite assessments of aerospace operators fit the journal's readership. Third, the paper addresses a research gap explicitly identified in two recent systematic reviews — Li, Molloy, El-Fiqi, & Eves (*Drones* 2025) for UAS operator cognitive-load assessment, and Rabat et al. (*BMJ Military Health* 2025) for warfighter mental endurance management — both of which called for integrated, inspectable frameworks that combine physiological monitoring with biomathematical fatigue estimation and task-specific calibration.

Contribution summary

The paper contributes four specific elements:

1. An explicit four-component weighted composite readiness index with per-task weight profiles derived from Multiple Resource Theory and related HF constructs.
2. A task taxonomy covering seventeen operator categories across manned aviation and UAS operations, with expected autonomic signatures and dominant failure modes.
3. Integration of a Warm-type vigilance-decrement model and a Chen-type control-latency penalty into a composite operator-readiness index — these components have no analogue in manned aviation and are absent from prior composite readiness frameworks.
4. An open-source reference implementation with MIT licensing, documented execution environment, and engineering verification across the fusion pathway.

The manuscript explicitly does **not** claim validated operator outcomes, diagnostic accuracy, numerical parity with external HRV packages, or regulatory clearance. Per-task weights are theory-derived pending field calibration. The illustrative worked example uses a single HRV recording and is framed as a framework-instantiation demonstration, not as inferential data. Limitations and a concrete validation roadmap (weight calibration, external HRV benchmarking, comparative benchmarking, prospective pilot study) are explicit in Section 4.

Suggested reviewers

(Optional; list to be finalised after internal review. Candidates familiar with the adjacent literature include researchers who have worked on composite pilot workload (Feng C., et al.; Military Psychology and NATO AFRL lineages including Stevens C. A., Morris M. B.), UAS operator cognitive-load assessment (Molloy O., Eves G.), cockpit physiological monitoring (Dehais F., ISAE-SUPAERO), and SAFTE/fatigue-management methodology (Hursh S. R., Devine J. K.). A formal list with institutional e-mails, ORCIDs, and diverse geography — per journal instructions — will be entered in Editorial Manager.)

Not under consideration elsewhere

This work is original and is not under consideration by any other journal. No portion has been previously published. **All authors** have approved the submission.

Data and code availability

Code, tables, and all manuscript support files are publicly accessible at `https://github.com/strike`
A tagged release or archived DOI corresponding to the exact reported version will be issued and cited in the final manuscript.

Conflicts of interest

The authors declare no competing financial or non-financial interests in relation to this manuscript. Neither author is serving or has served on the editorial board of *Applied Ergonomics*. The same declaration will be supplied on the submission portal via Elsevier's declaration-of-interest tool (Word upload) or the equivalent "no competing interests" confirmation, as required.

Thank you for considering this manuscript for *Applied Ergonomics*. I am available for any questions or clarifications at **diego.malpica@fac.mil.co**.

Sincerely,

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